

# TORSIONVERSE: CHEMICAL BONDS AND MOLECULAR MANIPULATION

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Companion script: analysis/demos/chemistry\_doc.py -- 44/44 PASS [BD1-BD15, JD1-JD12, CC1-CC12]

## ABSTRACT

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This paper derives four mechanisms for torsionverse-enhanced bond disruption and water splitting. (1) IRMPD: baseline sequential T<sub>1g</sub> excitation, 11.2 photons per O-H bond. (2a) A<sub>g</sub> orthogonal excitation at 90 deg:  $T_{1g} \times T_{1g} = A_g + T_{1g} + H_g$  [exact, I<sub>h</sub> CG] gives single-step direct excitation of the O-H symmetric stretch. (2b) C<sub>5</sub> enhancement at 72 deg:  $\chi(T_{1g}, C_5) = \phi$  gives  $\phi^2 = 2.618$ x more A<sub>g</sub> per unit intensity than 90 deg, reducing the effective photon count from 11.2 to 4.3 per bond [CC8]; possibly also includes I<sub>h</sub> medium coupling (open). (3) Pulsed oversize toroidal B-field: creates T<sub>2g</sub> vortex in the medium, depleting the Zone 2 pressure well and reducing the effective bond dissociation energy. (4) Photonic ring resonator / annular EM focus: circulating T<sub>1g</sub> TE<sub>01</sub> mode achieves milliwatt CW bond excitation [BD8-BD9] or pulsed annular focus with IRMPD window confirmed at R=1.5  $\mu$ m [BD12-BD13].

Section 5b derives the torsionverse mechanism for covalent bonding: the photon in a closed (1,2) trajectory has a 2-cell displacement budget (from linking number  $n = p \cdot q = 2$ , CC12), explaining the 2-center preference, Pauli exclusion, and electron degeneracy pressure from a single first principle.

The 72 deg /  $\phi^2$  prediction is falsifiable: two OPO lasers at 2.70  $\mu$ m; measure bond-breaking yield at 72 deg vs 90 deg; expected ratio =  $\phi^2$ .

NOTE ON PRIOR ART: General crossed-beam IR photodissociation is published (IRMPD, 2D-IR spectroscopy). The 72 deg /  $\phi^2$  prediction from I<sub>h</sub> group theory is new. No existing framework predicts the specific enhancement ratio or the group-theoretic origin of the optimal angle.

Magnetic pre-alignment (see Section 1b) enhances efficiency of Mechanisms 2a and 2b by orienting water's dipole moment (1.85 Debye) before the excitation pulse: all molecules present the O-H bond at the same angle, enabling coherent (not statistical) excitation. The magnetic pulse must be shorter than the O-H coherence time (sub-picosecond) to avoid decoherence while achieving bond-pressure alteration.

Cross-references:

|                  |                                                            |
|------------------|------------------------------------------------------------|
| doc_nucleus.txt  | -- Zone 1/2/3 structure, proton and atomic shells          |
| doc_electron.txt | -- Electron orbital forces, Bohr radius, Z <sub>crit</sub> |

## 1. BOND PHYSICS FROM TORSION MEDIUM GEOMETRY

### 1.1 The O-H bond as a torsion pressure well

Bond energy:  $E_{OH} = 5.15 \text{ eV}$  [NIST]  
Bond length:  $r_{OH} = 0.958 \text{ Angstrom} = 9.58 \times 10^{-11} \text{ m}$   
O-H stretch:  $\nu = 3700 \text{ cm}^{-1} \rightarrow h\nu = 0.4587 \text{ eV}, \lambda = 2.70 \text{ um}$   
Zone 2 pressure at bond scale:  $P = E_{OH} / V_{bond} = 224 \text{ GPa}$  [BD1 PASS]

The O-H bond is a Zone 2 pressure well at molecular scale. The same Zone 2 physics that holds quarks inside a proton (at  $r \sim \lambda_p = 0.21 \text{ fm}$ ) holds the O-H electrons between the oxygen and hydrogen nuclei (at  $r \sim 0.958 \text{ A}$ ). The scale difference is 4.5 orders of magnitude; the mechanism is identical.

### 1.2 Bond vibrational modes and $I_h$ symmetry

The O-H stretch frequency (111 THz) is the  $A_g$  mode of the O-H bond at molecular scale.  $A_g$  is the totally symmetric  $I_h$  irreducible representation -- an isotropic radial oscillation of the bond's Zone 2 pressure well (all directions simultaneously, no preferred axis). In standard spectroscopy this is the symmetric stretch ( $\nu_1$ ); in the torsionverse it IS the  $A_g$  mode of the local  $I_h$  pressure geometry.

TORSIONVERSE MECHANISM FOR  $\nu_1$ :

- (1)  $T_{1g}$  pressure at the bond locus drops isotropically ( $A_g$  reduction)  
-- the Zone 2 pressure well shallows uniformly in all directions.
- (2) The electron ( $E^+$  vertex mode) follows the pressure minimum outward by dwell-time asymmetry -- it drifts away from the bond centre.
- (3)  $T_{1g}$  pressure restores -- the well deepens again.
- (4) Electron rebounds to equilibrium.

Repeat at 111 THz. This is  $\nu_1$ : periodic isotropic pressure oscillation of the bond well, with the electron tracking the pressure gradient.

WHY  $A_g$  EXCITATION BREAKS THE BOND:

At sufficient amplitude, the pressure well drops enough that the electron drifts too far to rebound -- it escapes the Zone 2 pressure minimum and the bond dissociates. No collision or photon energy threshold required; the bond breaks when the isotropic pressure reduction exceeds the well depth.

CONNECTION TO CROSSED-BEAM MECHANISM (Section 3):

Two  $T_{1g}$  beams crossing at 90 deg create a local  $T_{1g}$  pressure reduction at the crossing point. This IS an  $A_g$  perturbation (isotropic, no preferred direction) -- it couples directly into the  $\nu_1$  oscillation without requiring the sequential ladder that IRMPD uses.

$A_g$  at cell scale is the Higgs/cell isotropic mode ( $E_{cell} = 124.8 \text{ GeV}$ );  
 $A_g$  at bond scale is the O-H symmetric stretch (0.46 eV). Same irrep, of an icosahedral elastic structure, differing only in the scale of the enclosing geometry.

H-O-H bend:  $\nu_2 = 1595 \text{ cm}^{-1} \rightarrow T_{1g}$  symmetry (directed compression)  
O-H stretch (symm):  $\nu_1 = 3657 \text{ cm}^{-1} \rightarrow A_g$  symmetry (isotropic)  
O-H stretch (asymm):  $\nu_3 = 3756 \text{ cm}^{-1} \rightarrow T_{1g}$  symmetry (directed)

Key: IRMPD targets the  $A_g$  mode ( $\nu_1$ ) because it builds amplitude without directionality. The asymmetric stretch ( $\nu_3, T_{1g}$ ) is less efficient for accumulating bond-breaking energy because its  $T_{1g}$  symmetry redirects energy along a specific axis rather than concentrating it in the bond.

### 1b. MAGNETIC PRE-ALIGNMENT (NEW INGREDIENT)

Water molecules have a permanent electric dipole moment:  $\mu = 1.85 \text{ Debye}$ . In an external magnetic or electric field, the molecular dipole aligns with the field. When aligned, every O-H bond in the sample presents at the same angle to the excitation beam -- coherent excitation becomes possible instead of the statistical averaging that reduces efficiency in unaligned samples.

PULSE REQUIREMENTS:

Alignment field: applied BEFORE the excitation pulse  
Duration: must be longer than the rotational relaxation time of water ( $\tau_{\text{rot}} \sim 8$  ps at 300 K) to achieve alignment, but the EXCITATION pulse must arrive BEFORE thermal disruption restores random orientation.  
Excitation pulse: sub-picosecond (shorter than O-H coherence time  $\sim 100$  fs) to achieve coherent A<sub>g</sub> excitation without decoherence from thermal noise.  
Sequence: align (ns field pulse) -> excite (fs laser pulse) -> measure

IN TORSIONVERSE TERMS: the external field creates a preferred orientation for the medium's local cell structure around each molecule. When the excitation beams arrive, every A<sub>g</sub> mode is in phase (all bonds at the same angle to the beam crossing point). The  $\phi^2$  enhancement from the 72-degree geometry is MULTIPLIED by the coherence enhancement from alignment (factor  $\sim N_{\text{aligned}}/N_{\text{total}} \rightarrow$  approaches 1.0 for strong alignment fields).

#### PRACTICAL GEOMETRY:

Field:  $\sim 1$ - $10$  T magnetic field or  $\sim 10^6$  V/m electric field (strong, pulsed)  
Laser: two OPO beams at 2.70  $\mu\text{m}$  (111 THz), crossed at 72 deg  
Target: water film or droplet at the field/beam crossing point  
Observable: bond-breaking yield vs angle (72 vs 90 vs 0 vs 180 deg)

## 2. MECHANISM 1: IRMPD SEQUENTIAL LADDER (EXISTING, SCRIPTED)

IR multiphoton dissociation: photons at 2.70  $\mu\text{m}$  resonance excite the O-H bond sequentially. Each absorbed photon climbs one rung of the vibrational ladder. N photons needed:

$$N = E_{\text{bond}} / h\nu = 5.15 \text{ eV} / 0.4587 \text{ eV} = 11.2 \text{ photons} \quad [\text{BD5 PASS}]$$

This is the TRUE multi-photon regime ( $N \gg 1$ ). Each photon must be absorbed before the next arrives; coherence is not required. The mechanism is:

1. Photon absorbed by O-H stretch mode (A<sub>g</sub>, 3700  $\text{cm}^{-1}$ )
2. Bond amplitude increases
3. Repeat N=11 times
4. Bond breaks at the dissociation threshold

#### SOURCE REQUIREMENTS:

Single-photon DUV path:  $\lambda < 240$  nm;  $E > 5.15$  eV per photon [BD4 PASS]  
IRMPD at 2.70  $\mu\text{m}$ : requires 11.2 photons,  $< 5$  ps pulse for coherence [BD5]  
Brute-force field:  $I > 10^{14}$  W/ $\text{cm}^2$  -- exceeds ionization threshold [BD2 PASS]  
Anti-Helmholtz static B: gradient force negligible ( $F/F_{\text{bond}} \sim 4e-17$ ) [BD6]  
Diamagnetic force:  $F_{\text{dia}} \sim 1e-3$  pN  $\ll F_{\text{bond}} = 8.6$  nN [BD7 PASS]

PRACTICAL APPROACH: CO<sub>2</sub> laser (10.6  $\mu\text{m}$ ) is NOT resonant with O-H stretch.  
Recommended: OPO source at 2.70  $\mu\text{m}$ , focused to  $I \sim 1e10$  W/ $\text{cm}^2$ .  
Ring cavity enhancement ( $Q \sim 1e5$ ) reduces required source power by  $10^5$ x.  
[BD8-BD15 PASS in bond\_disruption\_em.py]

MEDIUM SHEAR RESONANCE: the cell shear mode at 19.76 THz =  $R_s \cdot f_{\text{photon}}$  ( $R_s = \sqrt{5}/(4\pi)$ ) provides a secondary coupling channel directly to cell displacement. Combined O-H + shear drive:

$f_{\text{OH}} = 111$  THz (bond climbing)  
 $f_{\text{shear}} = 19.76$  THz (cell displacement)

Both can be driven simultaneously with a two-colour field. [JD12 PASS]

## 3. MECHANISM 2a: A<sub>g</sub> ORTHOGONAL-BEAM EXCITATION AT 90 DEG (NEW)

PRINCIPLE (bond resonance): Two T<sub>1g</sub> (EM) beams crossing at 90 degrees produce, via the I<sub>h</sub> CG decomposition:  $T_{1g} \times T_{1g} = A_g + T_{1g} + H_g$  [exact].  
The A<sub>g</sub> component IS the O-H symmetric stretch ( $\nu_1$ , 3700  $\text{cm}^{-1}$ ) -- the totally symmetric isotropic oscillation of the bond's Zone 2 pressure well.  
Direct single-step excitation of the A<sub>g</sub> mode -- not a sequential ladder.

#### ADVANTAGE OVER IRMPD:

IRMPD: 11.2 steps, sequential, requires sustained resonance over picoseconds.  
A<sub>g</sub> direct: one step, potentially much lower photon count per bond break.

#### BEAM CONFIGURATION:

Two beams crossing at 90 deg at the target bond. Photon energy: 0.4587 eV (2.70  $\mu\text{m}$  wavelength, same as IRMPD). A<sub>g</sub> is produced by the geometry alone -- no special angle precision beyond 90 deg required.

#### ELECTRON STRIPPING AS SIDE EFFECT:

If A<sub>g</sub> amplitude drives Jobson cells to Maxwell jamming (3V-E=6) at the bond location, Zone 3 co-rotation stops → electron stripped without photon energy threshold. [See doc\_electron.txt Section 2.1]

### 3a. MECHANISM 2b: 72-DEG C5 ENHANCEMENT -- I<sub>h</sub> MEDIUM COUPLING (NEW PREDICTION)

DISTINCT FROM MECHANISM 2a: at 72 deg (C5), a second effect may operate beyond simply producing more A<sub>g</sub>. Two interpretations, both open:

#### INTERPRETATION A (quantitative A<sub>g</sub> enhancement only):

$\chi(T_{1g}, C5) = \phi \rightarrow A_g \text{ yield} \propto \phi^2 = 2.618 \times$  vs 90 deg.  
The I<sub>h</sub> medium has a natural C5 mode; 72 deg excites it resonantly, producing  $\phi^2$  more A<sub>g</sub> per unit intensity. Same mechanism as 2a, stronger.  
Prediction: bond-breaking yield at 72 deg / 90 deg =  $\phi^2 = 2.618$ .  
Falsifiable: two OPO lasers at 2.70  $\mu\text{m}$ ; rotate crossing from 90 to 72 deg.

#### INTERPRETATION B (medium-level weakening added):

At 72 deg, the crossing also perturbs the I<sub>h</sub> medium at its C5 resonance. This is a MEDIUM COUPLING effect (analogous to Mechanism 3) triggered by EM geometry rather than a magnetic field. The bond sits in a medium that is simultaneously excited at C5; the medium pressure well supporting the bond is perturbed. This would reduce the effective bond dissociation energy ON TOP of the A<sub>g</sub> excitation -- a dual effect at 72 deg not present at 90 deg.  
OPEN: determine whether the  $\phi^2$  factor comes from A<sub>g</sub> amplitude alone (Interpretation A) or includes a medium-weakening component (Interpretation B).

NOTE:  $\chi(T_{1g}, \theta)$  increases monotonically as  $\theta$  decreases (smaller crossing angle = more A<sub>g</sub> by formula). But C6 (60 deg), C8 (45 deg) etc. are NOT elements of I<sub>h</sub> -- no medium resonance guarantees the enhancement there. 72 deg (C5) is the smallest I<sub>h</sub>-resonant angle with positive  $\chi$ . [CC1-CC4]

### 3b. MECHANISM 3: PULSED OVERSIZE TOROIDAL MAGNETIC FIELD -- MEDIUM DEPLETION

This mechanism does NOT work by coupling to the bond directly (diamagnetic gradient force is negligible:  $F_{\text{dia}}/F_{\text{bond}} \sim 1e-17$  [BD7]). It works by depleting the torsion medium at the toroid axis.

#### PHYSICAL SETUP:

An oversize toroidal electromagnet -- large enough for water to flow through the center hole -- is pulsed rapidly. The pulsed B-field (T<sub>2g</sub> mode in the torsionverse) creates a sustained T<sub>2g</sub> vortex in the torsion medium along the toroid axis. This depletes medium cells from the axis, lowering the local T<sub>1g</sub> (EM/pressure) density in the region where the water flows.

#### MECHANISM: MEDIUM DEPLETION, NOT BOND FORCE

The torsionverse bond model: O-H bonds are Zone 2 pressure wells sustained by the surrounding medium's T<sub>1g</sub> pressure field. The Zone 3 co-rotation of oxygen and hydrogen sets the dwell-time minimum that defines the bond. In a medium-depleted region (fewer cells → lower T<sub>1g</sub> pressure):

- Zone 3 orbital pressure gradient weakens
- The effective bond dissociation energy decreases
- Bonds that are thermally stable in full medium may dissociate spontaneously or break at significantly lower IRMPD fluence

This is the same principle as the spinning pyramid reducing apparent weight: centrifugal T<sub>1g</sub> depletion on the spin axis lowers effective gravity for objects there. Here, T<sub>2g</sub> vortex depletion at the toroid axis lowers effective bond stability for molecules there.

#### COMPARISON TO MECHANISM 2 (orthogonal beams):

Both mechanisms reduce the effective bond stability by modifying the medium.

- Mechanism 2: T<sub>1g</sub> orthogonal beams create an A<sub>g</sub> excitation that directly drives the bond's Zone 2 pressure well into A<sub>g</sub> oscillation from ABOVE (added energy into the bond)
- Mechanism 3: T<sub>2g</sub> toroidal vortex removes medium pressure support from BELOW (reduces the pressure well that the bond sits in)

The two are complementary and can be combined: weaken the bond via toroid, then break it with IRMPD or A<sub>g</sub> beams at a fraction of the normal fluence.

#### ADVANTAGE:

The toroid is azimuthally symmetric; no critical crossing angle.

Water flows continuously through the axis (scalable throughput).  
The depletion effect persists for the medium relaxation timescale after the pulse ends (not  $\tau_{\text{relax}} = 1\text{e8 s}$ ; this is elastic medium response, which relaxes on acoustic timescales when the driver stops).

OPEN: quantify the depletion depth (fractional  $T_{1g}$  pressure reduction) as a function of toroid pulse parameters (dB/dt, peak B, pulse width).  
This requires the  $T_{2g}$ /medium coupling coefficient (not yet derived).  
Compare to `jobson_depletion.py` (JD1-JD12) for the vortex depletion framework.

### 3c. MECHANISM 4: PHOTONIC RING RESONATOR / TOROIDAL EM FOCUS (NEW)

Distinct from Mechanism 3 (macroscopic magnetic, medium depletion) and Mechanism 2 (two separate beams crossing). Mechanism 4 uses a single circulating EM field in a toroidal geometry at MOLECULAR scale, adding energy directly to the bond -- not depleting the medium.

IMPLEMENTATION A -- Photonic ring resonator (BD8):  
A micron-scale ring cavity at  $2.7\ \mu\text{m}$  (chalcogenide glass, achievable  $Q \sim 1\text{e5}$ ) stores circulating EM energy at the O-H stretch resonance.  
Required input intensity:  $I_{\text{IRMPD}} / Q \sim 1\text{e4 W/cm}^2$  (milliwatt CW laser).  
Intracavity field continuously addresses the O-H bond at resonance.

IMPLEMENTATION B --  $T_{1g}$  vortex  $TE_{01}$  mode (BD9):  
The  $TE_{01}$  circulating mode of a ring resonator carries orbital angular momentum  $\ell=1$ , which corresponds to  $T_{1g}$  symmetry in  $I_h$ .  
 $\chi(T_{1g}, C5) = \phi \rightarrow \phi^2$  coupling to the  $A_g$  ( $\nu_1$ ) symmetric stretch.  
Combined with the  $Q=1\text{e5}$  cavity: required input  $< 1\text{e4} / \phi^2 \sim 4\text{e3 W/cm}^2$ .  
A 1 mW laser focused to  $1\ \mu\text{m}^2 = 1\text{e8 W/cm}^2 \rightarrow$  comfortably feasible.

IMPLEMENTATION C -- Above-cutoff annular laser focus (BD10-BD15):  
An annular beam (axicon + SLM) with inner radius  $R > R_{\text{cutoff}} = 1.04\ \mu\text{m}$ . Above cutoff, EM fields from all azimuthal positions converge inward and sum constructively at the central axis.  
 $I_{\text{center}} \approx 0.5 \times I_{\text{wall}} \times \phi^2$ ; at  $I_{\text{wall}} = 1\text{e10 W/cm}^2$ :  $I_{\text{center}} = 1.3\text{e10 W/cm}^2$ , inside the IRMPD window without ionization [BD12-BD13].  
Practical aperture:  $1.04\ \mu\text{m} < R_{\text{inner}} < \sim 10\ \mu\text{m}$  [BD15].  
Water flows along center axis (microfluidic integration).

KEY DISTINCTION from Mechanism 2:  
Mechanism 2: two separate beams cross  $\rightarrow A_g$  at intersection (point geometry)  
Mechanism 4: single circulating mode wraps around axis  $\rightarrow$  continuous  $T_{1g}$  along the axis (line geometry, better suited for flowing water)

Companion script: `bond_disruption_em.py` BD8-BD15 (all verified)

## 4. WATER SPLITTING: MECHANISM COMPARISON

REACTION:  $2\text{ H}_2\text{O} \rightarrow 2\text{ H}_2 + \text{O}_2$  ( $\Delta G = 237\text{ kJ/mol} = 2.46\text{ eV/molecule}$ )  
Two O-H bonds per molecule must be broken: total  $\sim 10.3\text{ eV}$  per molecule.

MECHANISM 1 (IRMPD):  
 $N = 11.2$  photons per O-H bond  $\times 2$  bonds = 22.4 photons per molecule.  
Energy per molecule =  $22.4 \times 0.4587\text{ eV} = 10.3\text{ eV}$  [consistent].  
All energy goes to bond excitation; product gases carry kinetic energy.

MECHANISM 2a ( $A_g$  direct,  $90^\circ$  deg orthogonal beams):  
Geometry produces  $A_g$  directly. One step vs 11.2 for IRMPD.  
No special angle precision required;  $90^\circ$  deg is the base configuration.  
OPEN: compute  $A_g$  coupling efficiency at the O-H bond geometry.

MECHANISM 2b ( $C5$  enhancement,  $72^\circ$  deg):  
 $\phi^2 = 2.618\times$  more  $A_g$  per unit intensity than  $90^\circ$  deg.  
Effective photon count:  $N = 11.2 / \phi^2 = 4.3$  per bond [CC8].  
Interp. A: more  $A_g$  amplitude (same mechanism, stronger).  
Interp. B:  $I_h$  medium coupling adds a medium-weakening component.  
OPEN: determine which interpretation is correct.  
Falsifiable: measure bond-breaking yield at  $72^\circ$  deg vs  $90^\circ$  deg; expect  $\phi^2$ .

MECHANISM 3 (toroidal B-field -- medium depletion):  
 $T_{2g}$  vortex reduces medium pressure at toroid axis.

Effective bond dissociation energy decreases in the depleted zone.  
 Best used in combination: weaken bonds via toroid, then break with IRMPD or A<sub>g</sub> beams at reduced fluence. Azimuthally symmetric; continuous flow.  
 OPEN: quantify depletion depth from pulse parameters (T<sub>2g</sub> coupling TBD).

MECHANISM 4 (photonic ring / toroidal EM focus):

Ring resonator (BD8): input  $\sim 1\text{e}4\text{ W/cm}^2$  (milliwatt CW,  $Q=1\text{e}5$ ). [PASS BD8]

TE<sub>01</sub> + T<sub>1g</sub>  $\phi^2$ : input  $< 4\text{e}3\text{ W/cm}^2$ . [PASS BD9]

Above-cutoff annular focus (BD10-BD15):  $I_{\text{wall}} = 1\text{e}10\text{ W/cm}^2 \rightarrow$

$I_{\text{center}} = 1.3\text{e}10\text{ W/cm}^2$  in IRMPD window; aperture 1-10  $\mu\text{m}$ .

Best for continuous flow; line geometry vs point geometry of Mechanism 2.

DOWNSTREAM (all mechanisms):

Products H<sub>2</sub> and O<sub>2</sub> must be separated (H<sub>2</sub> rises, O<sub>2</sub> denser).

Electrolyzer cells provide existing engineering path for separation.

Torsionverse mechanisms would reduce the ENERGY INPUT, not the downstream.

## 5. OTHER BOND TARGETS

C-H bond:  $E = 4.28\text{ eV}$ ,  $\nu = 3000\text{ cm}^{-1}$  (3.33  $\mu\text{m}$ ) [hydrocarbons]

N-H bond:  $E = 3.88\text{ eV}$ ,  $\nu = 3300\text{ cm}^{-1}$  (3.03  $\mu\text{m}$ ) [amines, proteins]

C-C bond:  $E = 3.61\text{ eV}$ ,  $\nu = 1000\text{ cm}^{-1}$  (10.0  $\mu\text{m}$ ) [organic chemistry]

Si-O bond:  $E = 4.60\text{ eV}$ ,  $\nu = 1000\text{ cm}^{-1}$  (10.0  $\mu\text{m}$ ) [silica, glass]

The A<sub>g</sub> orthogonal-beam mechanism applies to any bond whose vibrational A<sub>g</sub> mode is accessible. The key requirement: the T<sub>1g</sub> beam photon energy must match the bond's A<sub>g</sub> stretch frequency. For C-H bonds: 3.33  $\mu\text{m}$  OPO source.

SELECTIVITY ADVANTAGE: unlike IRMPD, which requires a well-resolved spectrum, the A<sub>g</sub> direct mechanism selects for the symmetric Zone 2 oscillation (A<sub>g</sub> symmetry). In a complex molecule, only the bonds whose A<sub>g</sub> mode frequency matches the beam photon energy will be excited -- inherent selectivity.

## 5b. THE COVALENT BOND: WHY ELECTRON SHARING IS STABLE

Standard quantum mechanics resolves covalent bonding via wavefunction overlap and exchange integrals, but provides no mechanical picture. The torsionverse gives a direct mechanical explanation from a single fundamental object.

THE FUNDAMENTAL OBJECT: THE PHOTON

In the torsionverse there is one fundamental moving object: the photon. What we call the "electron" is a photon in a stable closed trajectory through the Jobson cell lattice. The (1,2) topology is NOT an intrinsic property of the photon -- it is a property of the TRAJECTORY through the icosahedral cell geometry. The photon bounces in straight lines between cells; the icosahedral lattice forces specific stable closed trajectories; the (1,2) trajectory is one such stable loop.

THE 2-CELL BUDGET (from trajectory geometry):

Linking number  $n = p \cdot q = 2$  counts simultaneous transverse crossings. [CC12a]  
 Dynamic picture: the electron ricochets off a DIFFERENT cell at each bounce with an elastic rebound -- the cell displaces and recovers under global medium pressure. The electron travels to the next cell while the previous cell is still displaced. If each cell is displaced for 2 bounce intervals ( $\tau_{\text{recover}} \sim T/6$ , where  $T/12$  is the inter-bounce interval), then at any moment  $\sim 2$  cells are simultaneously still displaced from recent impacts. Both the topological count ( $n=2$  crossings) and the dynamic picture (2 cells displaced) give the same budget: 2 cells.

Over a complete cycle: 12 distinct cells visited, 6 sequential 2-cell pairs.

[CC12b, CC12c]. With vertex stiffness correction:  $n_{\text{exact}} = 2.018697$ .

For all bond discussions: the budget is 2 cells.

ENERGY BUDGET (2-cell budget):

The photon's trajectory uses its 2-cell budget:

$\sim 1$  cell: the trajectory presses into the nucleus Zone 3 cell geometry (the nuclear charge well is a specific region of the cell lattice)

$\sim 1$  cell: the trajectory's orbital motion + Pauli repulsion capacity

PAULI EXCLUSION: a second photon on the same trajectory would double the cell pressure  $\rightarrow$  their trajectories repel each other unless the second photon circles the same path in the anti-parallel orientation (same cells visited, winding axis reversed). Anti-parallel: the two T<sub>1g</sub> circulations cancel via

A<sub>g</sub> decoupling ( $\chi(T_{1g\_up} \times T_{1g\_down}, C_2) = +1 = A_g$ ) → coexist.  
 Parallel: circulations reinforce → repulsion → excluded.  
 NOTE: opposite winding chirality = POSITRON (same cells, reversed order = anti-particle). Pauli pairs are NOT positrons -- they have anti-parallel axes, not reversed chirality.

#### COVALENT BOND (two-center):

When a second nucleus is close enough, the photon's closed trajectory can include cells from BOTH nuclear lattice regions -- alternating between nucleus A's Zone 3 cells and nucleus B's Zone 3 cells. The trajectory splits its 2-cell budget: ~1 cell in A's region, ~1 cell in B's region. Zero budget remains for orbital motion or Pauli repulsion. The trajectory is LOCKED to the path that spans both nuclei. This is the bond.

#### WHY EXACTLY 2 CENTERS (not 3):

The 2-cell budget supports exactly 2 nuclear cell regions. A trajectory spanning 3 nuclei would need  $n > 2$  -- no stable (p,q) winding achieves this with  $p \cdot q = 2$ . The 2-center preference is a direct consequence of the same topology that fixes  $\alpha$  ( $n \cdot \alpha^2 - Q \cdot \alpha + R_s = 0$ ).

#### RESOLUTION OF THE "OVERLY POSITIVE" PROBLEM:

There is no charge redistribution. The photon's trajectory spans two nuclear cell regions; both nuclei are attracted toward the region where their cell geometries overlap (the trajectory's Zone 3 path through both). The Coulomb repulsion between bare nuclei at H-H bond distance is ~19.5 eV [CC11], but the Zone 3 overlap attraction is larger; net bond energy 4.52 eV is the residual.

#### PAULI EXCLUSION AND CELL DEPLETION:

Pauli exclusion works because the photon trajectory requires spare cell budget to repel a co-entering photon. In a depleted medium (fewer Jobson cells), the spare budget falls → exclusion fails → trajectories can overlap. This is electron degeneracy pressure: at high stellar density, cell depletion removes the spare budget → Pauli repulsion fails → collapse.  
 [OPEN: derive threshold cell density from  $n_{exact}$  and Zone 3 geometry]

#### BOND GEOMETRY:

Preferred trajectory directions in the icosahedral cell lattice → sp<sup>3</sup> tetrahedral bond geometry. H-O-H at 104.5 deg: the two locked photon trajectories (O-H bonds) weakly repel via residual spare budget from the boundary of their cell regions, compressing 5 deg below tetrahedral.

Script checks: CC10-CC12 in analysis/molecular/chemistry\_coupling.py

#### NOTE ON TWO PICTURES OF THE 2-CELL BUDGET:

TOPOLOGICAL (exact): the (1,2) trajectory simultaneously crosses any transverse disk at 2 points. Count is exactly 2 everywhere on the path.  
 SEQUENTIAL-DISPLACEMENT (dynamic): if cell recovery time  $\tau_{recover} = T/6$ , then  $12 \times \tau_{recover}/T = 2$  cells displaced at any moment.  
 Both give 2; whether  $\tau_{recover} = T/6$  is a medium property (open).

#### WAVE-PARTICLE NOTE:

There is only ONE fundamental object: the photon. Both the "electron" and the "free photon" are the same object in different trajectory states (closed loop vs open path). The apparent "wave" for both is the same: the photon's path through the Jobson cell lattice creates periodic displacement patterns via elastic rebounds. No separate wave exists; the pattern IS the trajectory.

## 6. GENUINELY OPEN ITEMS

- [CLOSED CC1-CC4] A<sub>g</sub> crossing angle: 72 deg (C<sub>5</sub>, I<sub>h</sub> resonant) gives  $\phi^2 = 2.618$  vs 90 deg. Non-I<sub>h</sub> angles (60, 45 deg) give higher  $\chi^2$  by formula but lack I<sub>h</sub> medium resonance --  $\phi^2$  is physically guaranteed only at 72 deg.
- [CLOSED CC5-CC7] T<sub>2g</sub>  $\chi$  table:  $\chi(T_{2g}, C_5) = 1 - \phi$ ; maximum  $|\chi(T_{2g})|$  is at C<sub>5</sub><sup>2</sup> (144 deg) =  $\phi$ , not C<sub>2</sub>. For water C<sub>2v</sub>, best is C<sub>2</sub> axis,  $|\chi| = 1$ .
- [BD7 noted] Diamagnetic gradient force (macroscopic B on molecule) ~ 1e-17 of bond force -- negligible. Mechanism 3 operates via medium depletion, not gradient force; BD7 measures a different (irrelevant) coupling path.
- Depletion depth of Mechanism 3: derive T<sub>2g</sub>/medium coupling vs pulse parameters (peak B, dB/dt, pulse width); compare to JD1-JD12 (jobson\_depletion.py)
- Threshold intensity comparison: IRMPD vs A<sub>g</sub> direct vs toroid-primed IRMPD
- Optimal combination: what toroid depletion depth reduces IRMPD fluence by  $\phi^2$ ?

- Electron stripping threshold: at what  $A_g$  amplitude does Maxwell jamming occur, causing cell-locking ionization?
- Plasma efficiency: in a water plasma, do any mechanisms scale differently than in liquid/vapor water?

## 7. PROPOSED EXPERIMENT

See experiments/bench\_test\_orthogonal\_waves.txt for full protocol.  
Three setups: (A) piezo + photoelastic acrylic block -- direct mode identification via polarized light; (B) ultrasound in water bath -- cavitation threshold comparison; (C) microwave patch antennas -- EM  $T_{1g}$  analog. Key discriminator: 72 deg crossing (C5 icosahedral angle) vs 90 deg. Torsionverse predicts  $\phi^2 = 2.618\times$  more  $A_g$  at 72 deg.